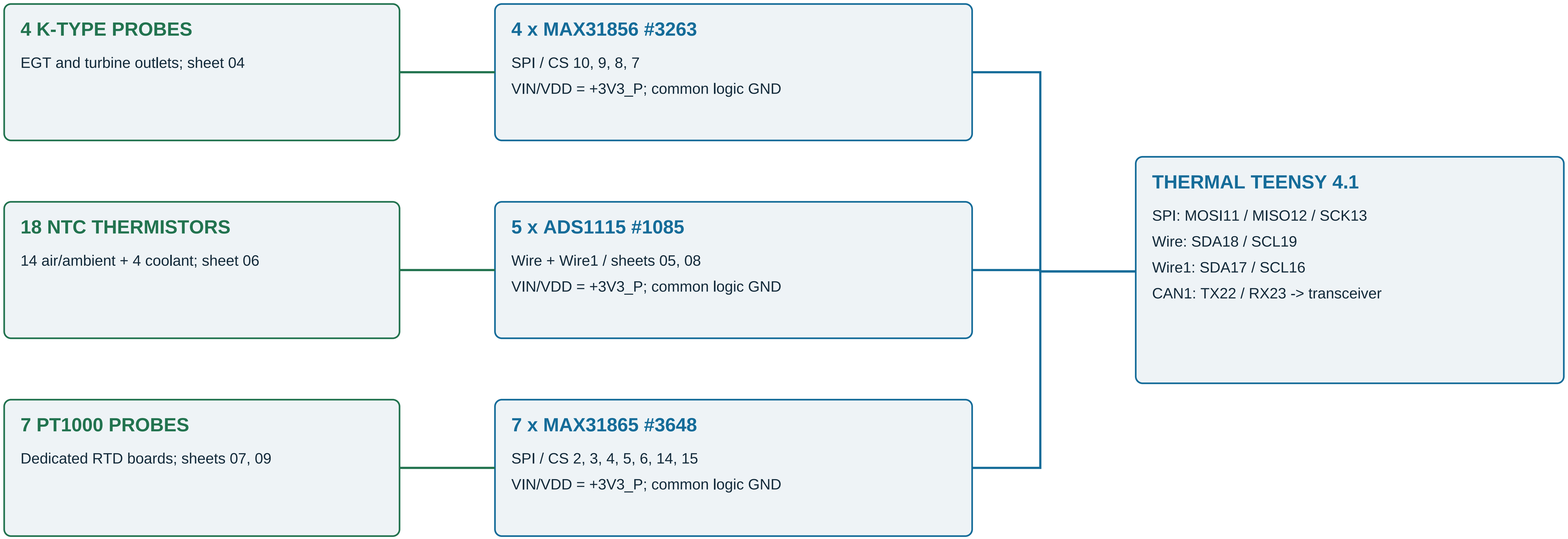
Recommended breakout-board architecture

29 active sensors / second Teensy 4.1 / Rev B supersedes the ADS7953 wiring plan



RETAINED ECU PROTECTION

Dedicated ECU coolant, IAT and oil sensors remain separate.

No parallel sensor inputs or ECU replacement signal in this plan.

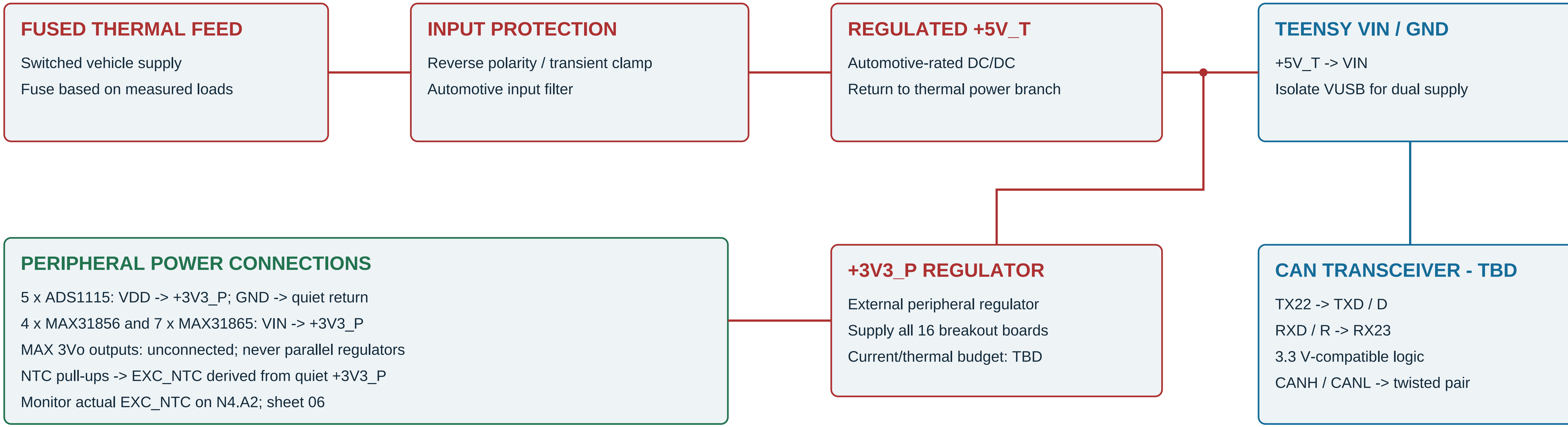
FIRMWARE IMPLEMENTED / BENCH HOLD

Thermal config 2.0.0 implements these breakout boards.

Use the matching new binary; validate wiring on the bench.

Protected power, grounds and CAN

No external ADS7953 reference, mux buffer or custom RTD current source in Rev B



GROUND, HARNESS AND POWER RULES

NTC B wires return to SGND at acquisition board; do not use engine metal as the sensor return.

PT1000 wires go only to their MAX31865 sensor terminals. Do NOT ground an RTD lead externally.

Bond quiet signal return and logic/power ground deliberately; keep pump/valve currents out of this path.

Shields terminate at a planned enclosure/chassis bond, not at TC minus or RTD minus. Bond detail remains TBD.

CAN is 500 kbit/s. Fit 120 ohm across H/L only at physical bus ends; verify common-reference offsets.

Keep SPI/I2C inside enclosure. Check rail sequencing; prevent back-power through unpowered digital inputs.

Power budget includes regulator losses, breakout bias/LED loads, transceiver and all NTC excitation currents.

SPI backbone and complete Teensy pin allocation

Physical thermal-node pins only; main-controller pins are a separate namespace

THERMAL TEENSY

11 -> all SDI
12 <- all SDO
13 -> all SCK
Common logic ground

4 x MAX31856

TC0..TC3 / Adafruit #3263
CS pins 10, 9, 8, 7
FLT / DRDY / 3Vo: unconnected

7 x MAX31865

R0..R6 / PT1000 #3648
CS pins 2, 3, 4, 5, 6, 14, 15
RDY / 3Vo: unconnected

Function	Teensy pins	Destination
SPI MOSI / MISO / SCK	11 / 12 / 13	All eleven SPI front ends
Thermocouple CS	10 / 9 / 8 / 7	TC0 / TC1 / TC2 / TC3
RTD CS	2 / 3 / 4 / 5 / 6 / 14 / 15	R0 / R1 / R2 / R3 / R4 / R5 / R6
Wire SDA / SCL	18 / 19	N0, N1, N2 only
Wire1 SDA / SCL	17 / 16	N3, N4 only
CAN1 TX / RX	22 / 23	Via selected transceiver; never directly to CANH/L

Each CS is active low; pull up to +3V3_P (proposed 10 kOhm, accounting for board pull-ups).

Initialize every CS high. Per-device SPI settings and high-impedance unselected SDO require bench verification.

Pins 5/6 are now RTD selects, NOT the old ADS7953 selects. Older ADS7953 binaries must not operate this harness.

Thermocouple harnesses and breakout boards

Four independent K-type pairs; proposed example breakout: Adafruit MAX31856 #3263, not a generic pin-order guarantee

T01 / EGT_LEFT

left exhaust pre-turbine

K-type probe; insulated junction preferred

K+ -> T+

K- -> T-

TC0 / MAX31856

T+ = positive / T- = negative

CS = Teensy 10

VIN <- +3V3_P; GND <- logic GND

SDI 11 / SDO 12 / SCK 13

3Vo, FLT and DRDY: leave unconnected

T02 / EGT_RIGHT

right exhaust pre-turbine

K-type probe; insulated junction preferred

K+ -> T+

K- -> T-

TC1 / MAX31856

T+ = positive / T- = negative

CS = Teensy 9

VIN <- +3V3_P; GND <- logic GND

SDI 11 / SDO 12 / SCK 13

3Vo, FLT and DRDY: leave unconnected

T03 / TURBINE_OUT_LEFT

left turbine outlet

K-type probe; insulated junction preferred

K+ -> T+

K- -> T-

TC2 / MAX31856

T+ = positive / T- = negative

CS = Teensy 8

VIN <- +3V3_P; GND <- logic GND

SDI 11 / SDO 12 / SCK 13

3Vo, FLT and DRDY: leave unconnected

T04 / TURBINE_OUT_RIGHT

right turbine outlet

K-type probe; insulated junction preferred

K+ -> T+

K- -> T-

TC3 / MAX31856

T+ = positive / T- = negative

CS = Teensy 7

VIN <- +3V3_P; GND <- logic GND

SDI 11 / SDO 12 / SCK 13

3Vo, FLT and DRDY: leave unconnected

THERMOCOUPLE ROUTING IS NOT ORDINARY COPPER SENSOR WIRING

Keep K-alloy extension cable and matched polarity connectors through to the cold-junction terminals. Do not tie T- to chassis or SGND. Confirm grounded-junction compatibility and input common-mode limits before using a grounded probe.

Keep boards away from heat gradients. These inputs are not galvanically isolated. MAX31856 source: sheet 10.

ADS1115 bus wiring and address straps

Five Adafruit #1085 boards / 20 single-ended inputs / no external I2C multiplexer

WIRE BUS

SDA18 / SCL19

Both lines pulled to +3V3_P

Do not join buses together

N0 / ADS1115

Address 0x48 / ADDR -> GND

VDD -> +3V3_P / GND -> SGND

ALRT: unconnected

N1 / ADS1115

Address 0x49 / ADDR -> VDD

VDD -> +3V3_P / GND -> SGND

ALRT: unconnected

N2 / ADS1115

Address 0x4A / ADDR -> SDA

VDD -> +3V3_P / GND -> SGND

ALRT: unconnected

WIRE1 BUS

SDA17 / SCL16

Both lines pulled to +3V3_P

Do not join buses together

N3 / ADS1115

Address 0x48 / ADDR -> GND

VDD -> +3V3_P / GND -> SGND

ALRT: unconnected

N4 / ADS1115

Address 0x49 / ADDR -> VDD

VDD -> +3V3_P / GND -> SGND

ALRT: unconnected

ADDRESS AND ELECTRICAL CHECKS

ADDR straps are named nets, not connector cavity numbers. N2 ADDR connects to its own SDA net.

Verify the purchased board revision/default ADDR link; do not leave a conflicting strap to ground.

Expected scan: Wire = 0x48, 0x49, 0x4A; Wire1 = 0x48, 0x49. Duplicate addresses across buses are intentional.

Account for parallel onboard SDA/SCL pull-ups. Measure effective resistance and rise time before choosing bus speed.

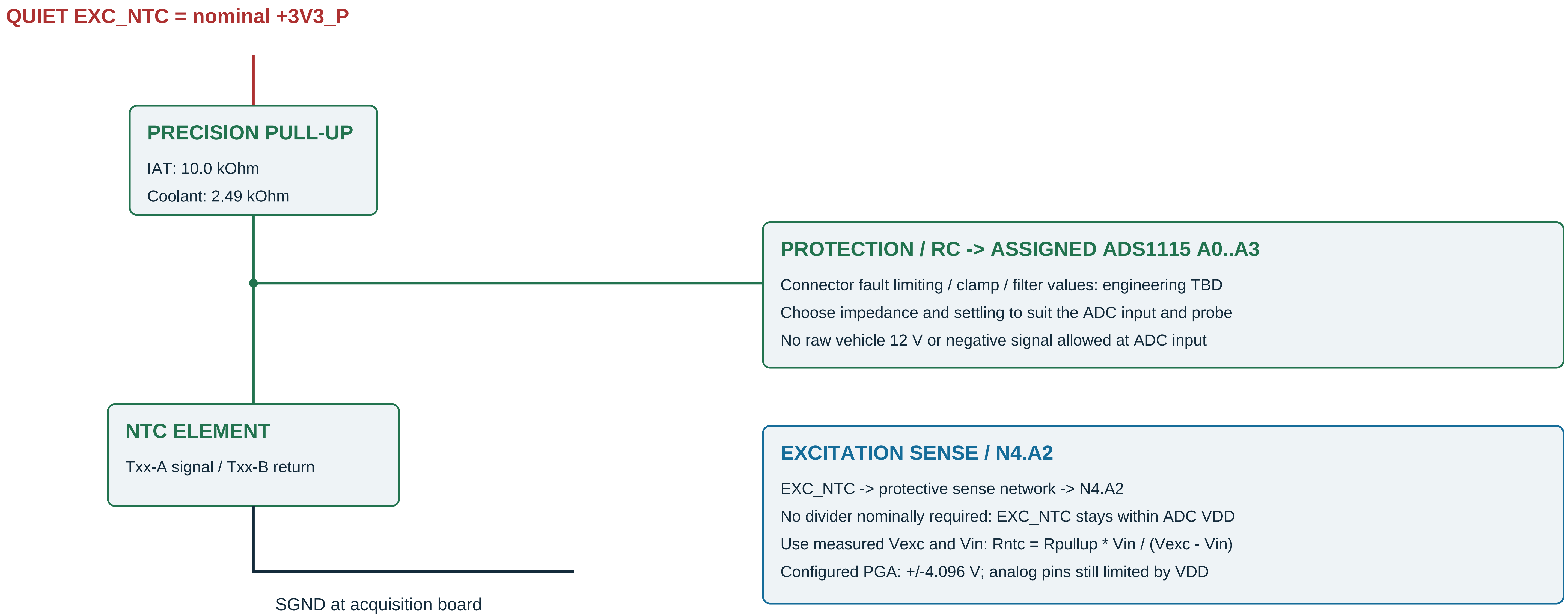
Do not add extra pull-ups blindly. Keep every I2C pull-up at 3.3 V, never 5 V.

Use a nonblocking conversion scheduler: the ADS1115 conversion rate is shared across its selected inputs.

Separate buses improve fault containment but are not galvanic isolation. A failed board invalidates its assigned channels.

Thermistor input circuit and excitation monitoring

Repeat for 18 probes; ADS1115 internal reference is NOT the thermistor excitation supply



SIGNED CONVERSION IMPLEMENTED / PROBE CALIBRATION REQUIRED

Firmware uses signed ADS1115 counts and measured N4.A2 excitation. Invalid excitation removes all NTC readings. Short/open and stale checks precede conversion; actual probe curves and settling still need validation.

Existing beta profiles are provisional: IAT R25=10k/B3435; coolant R80=2.50k/B3977. Verify actual probe tables.

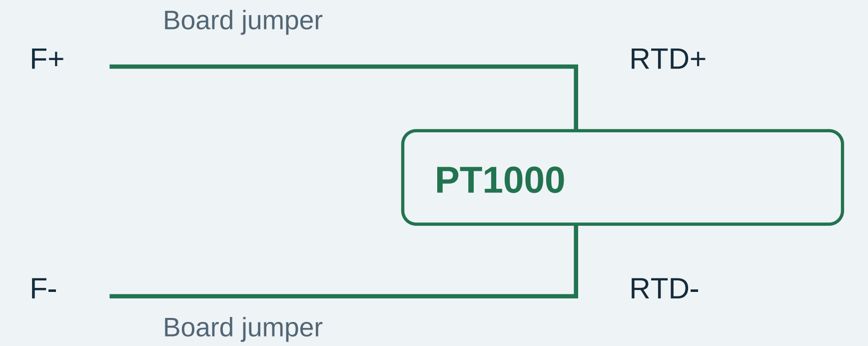
PT1000 wiring - MAX31865 replaces current sources

Selected board: Adafruit #3648 PT1000 version, with 4300 ohm reference resistor; NOT the PT100 version

TWO-WIRE BASELINE

- Retains the previous two-wire harness
- One isolated PT1000 per board
- Lead A -> RTD+; lead B -> RTD-
- Bridge F+ to RTD+ on the board
- Bridge F- to RTD- on the board

MAX31865 SENSOR TERMINALS



LOGIC / POWER

- VIN -> +3V3_P
- GND -> logic GND
- SDI11 / SDO12 / SCK13
- Unique CS: sheet 09
- RDY / 3Vo unused

OPTIONAL FOUR-WIRE HARNESS

- Preferred where probe packaging supports it.
- Two independent leads from element end A -> F+ and RTD+.
- Two independent leads from end B -> F- and RTD-.
- Use the board default four-wire jumper arrangement.
- No sense/force shorts at the converter for four-wire operation.
- Requires four conductors and a matching connector schedule.

THREE-WIRE ALTERNATIVE

- Equal-resistance paired leads -> F+ and RTD+.
- Single opposite-end lead -> RTD-; bridge F- to RTD-.
- Set the board-specific three-wire selector per Adafruit guide.
- Configure MAX31865 three-wire compensation in firmware.
- Do not infer wire function solely from insulation color.
- Use matching lead lengths/gauges for compensation.

DO NOT CONNECT RTD MINUS TO SGND OR ADD THE OLD 500 uA EXCITATION

Sensor current and resistance-ratio measurement belong to the MAX31865. Firmware uses R0=1000 ohm, Rref=4300 ohm, configured lead mode, fault detection and CVD conversion. Two-wire lead error remains.

Baseline CSV uses two wires. Four-/three-wire choices require explicit harness revision; shields are separate conductors.

NTC channel-by-channel harness schedule

Stable sensor IDs retained / board names N0-N4 are new Rev B assignments

ID	Sensor	Board	I2C bus/address	Input	Pull-up	Harness signal / SGND return
5	COMP_IN_LEFT	N0	Wire 0x48	A0	10k	T05-A / T05-B
6	COMP_IN_RIGHT	N0	Wire 0x48	A1	10k	T06-A / T06-B
7	COMP_OUT_LEFT	N0	Wire 0x48	A2	10k	T07-A / T07-B
8	COMP_OUT_RIGHT	N0	Wire 0x48	A3	10k	T08-A / T08-B
9	IC_IN_LEFT	N1	Wire 0x49	A0	10k	T09-A / T09-B
10	IC_IN_RIGHT	N1	Wire 0x49	A1	10k	T10-A / T10-B
11	IC_OUT_LEFT	N1	Wire 0x49	A2	10k	T11-A / T11-B
12	IC_OUT_RIGHT	N1	Wire 0x49	A3	10k	T12-A / T12-B
13	PRE_WMI	N2	Wire 0x4A	A0	10k	T13-A / T13-B
14	POST_WMI	N2	Wire 0x4A	A1	10k	T14-A / T14-B
15	PLENUM_IAT	N2	Wire 0x4A	A2	10k	T15-A / T15-B
16	RUNNER_IAT_LEFT	N2	Wire 0x4A	A3	10k	T16-A / T16-B
17	RUNNER_IAT_RIGHT	N3	Wire1 0x48	A0	10k	T17-A / T17-B
18	HEAD_COOLANT_LEFT	N3	Wire1 0x48	A1	2.49k	T18-A / T18-B
19	HEAD_COOLANT_RIGHT	N3	Wire1 0x48	A2	2.49k	T19-A / T19-B
22	RAD_IN	N3	Wire1 0x48	A3	2.49k	T22-A / T22-B
23	RAD_OUT	N4	Wire1 0x49	A0	2.49k	T23-A / T23-B
29	AMBIENT_AIR	N4	Wire1 0x49	A1	10k	T29-A / T29-B

N4.A2 = EXC_NTC sense (internal wire, no new temperature ID). N4.A3 = spare, unconnected and not sampled.

A/B are harness net names, not vendor pin numbers. Protect and filter every exposed NTC signal at enclosure entry.

RTD harness, unused inputs and board quantities

Seven independent converters / no PT1000 connected to any ADS1115 channel

ID	Sensor	Board	CS pin	Element end A	Element end B
20	HEAD_METAL_LEFT	R0	2	T20-A -> RTD+	T20-B -> RTD-
21	HEAD_METAL_RIGHT	R1	3	T21-A -> RTD+	T21-B -> RTD-
24	OIL_GALLERY	R2	4	T24-A -> RTD+	T24-B -> RTD-
25	OIL_COOLER_IN	R3	5	T25-A -> RTD+	T25-B -> RTD-
26	OIL_COOLER_OUT	R4	6	T26-A -> RTD+	T26-B -> RTD-
27	TURBO_OIL_DRAIN_LEFT	R5	14	T27-A -> RTD+	T27-B -> RTD-
28	TURBO_OIL_DRAIN_RIGHT	R6	15	T28-A -> RTD+	T28-B -> RTD-

BASELINE BOARD ORDER

5 x Adafruit ADS1115 #1085
7 x Adafruit MAX31865 PT1000 #3648
4 x Adafruit MAX31856 #3263 (retained)
1 x dedicated Teensy 4.1 (retained)
1 x 3.3 V-compatible CAN transceiver (TBD)
Power/NTC conditioning/connector hardware: still required

RESERVED IS NOT FITTED

IDs 30/31: CHRA PT1000 probes remain disabled.
No R7/R8 boards or CS pins allocated for these probes.
ID32: reserved, no physical input.
N4.A3 spare is not a direct PT1000 expansion input.
Two-wire baseline: install both force/sense bridges.
RTD B wire is NOT an SGND harness connection.

Purchase links and manufacturer pin/jumper references are listed on sheet 10 and in docs/thermal_wiring.md.

Commissioning boundary and source references

Acquisition implemented / wiring and sensor calibration still require physical bench validation

IMPLEMENTED ACQUISITION / CONFIG 2.0.0

Two-bus ADS1115: 100 kHz I2C / +/-4.096 V / 860 SPS.
Seven MAX31865: 10 ms bias + fault cycle + 66 ms wait.
New channel/CS map; all eleven selects initialize high.
NTC signed conversion + measured N4.A2 excitation.
PT1000 ratio, CVD equation and front-end fault decoding.
750 ms stale boundary; two fresh samples for recovery.
TC asynchronous single shots / requested 5 Hz cadence.
Native fault tests + Teensy compile; bench tests still required.

BENCH AND HARNESS CHECKS

Check supply polarity, regulator loading and USB/VIN isolation.
Verify every address and CS with one board at a time.
Check each probe identity using known resistances/TC simulator.
Check N4.A2 excitation reading against a calibrated meter.
Test open, short, board disconnect and stuck-bus behavior.
Check simultaneous sampling load and maximum channel age.
Verify shield bonding, transient protection and fault current paths.
Keep independent ECU temperature sensors connected.

ADS1115 board / purchase	https://www.adafruit.com/product/1085
ADS1115 electrical / conversion limits	https://www.ti.com/lit/ds/symlink/ads1115.pdf
MAX31865 PT1000 board / purchase	https://www.adafruit.com/product/3648
MAX31865 terminal jumpers	https://learn.adafruit.com/adafruit-max31865-rtd-pt100-amplifier/rtd-wiring-config
MAX31865 logic pinouts	https://learn.adafruit.com/adafruit-max31865-rtd-pt100-amplifier/pinouts
Teensy I2C ports / pins	https://www.pjrc.com/teensy/td_libs_Wire.html
MAX31856 logic and sensor terminals	https://learn.adafruit.com/adafruit-max31856-thermocouple-amplifier/pinouts
Teensy pin card / CAN1	https://www.pjrc.com/teensy/card11a_rev4_web.pdf

Exact probe curves, connector cavities, NTC protection/RC values, power parts and CAN module are still engineering holds.